Channel Estimation for Rydberg Receivers Using Learned Hankel Filtering

Abdullah Haatim

***Abstract*—Rydberg receivers use excited atoms to sense radio signals. Magnitude-only observations do not directly reveal the wireless channel. Channel structure can help recover it, but fixed filtering may remove useful details. We study a small learned estimator that alternates measurement fitting with Hankel filtering. It learns from withheld observations, without true-channel targets. Simulations show lower average error and processing time than the tested classical Hankel estimator. Both structure and noise information help average accuracy. However, the classical method remains better with a weak reference and in some few-path channels. Learned filtering is a useful option for the tested model within these operating limits.**

***Index Terms*—Channel estimation, Hankel filtering, magnitude measurements, Rydberg receivers.**

I. Introduction

RYDBERG atoms respond to radio-frequency electric fields, offering a way to study new wireless receivers. Here, signals from several users combine with a known reference. The receiver measures their combined magnitude, not their full complex value. The channel describes changes in signal strength and phase on the way to the receiver. Model-based methods and learned methods such as URformer estimate it [1], [2].

A few propagation paths can create a repeated pattern across a regular array. Arranging channel samples in a Hankel matrix lets an estimator use this pattern. Strong filtering can remove real details; weak filtering can leave noise. Searching many filtering settings also takes time. We study a small model that learns how much filtering to apply. The true channel is never shown to it during training. Instead, some measurements are hidden from the estimator and used to check its predictions. This is called self-supervised learning.

Our contributions are: (i) a 188-parameter estimator that combines Rician measurement updates with smooth Hankel filtering; (ii) training this estimator on withheld measurements, without channel targets or channel-error checkpoint selection; and (iii) paired tests of accuracy, processing time and simpler versions, including fresh-test replication and input-removal tests. We report weak-reference and few-path failures alongside the gains.

II. Related Work

Training on withheld measurements is established in image reconstruction [3] and self-supervised denoising is used for Rydberg sensor signals [4]. Huang et al. learn Hankel-factor updates from fully sampled targets [5]. Luo et al. combine self-supervision and structured low-rank reconstruction for complex MRI data [6]. DeepTensor fits network-based low-rank factors to an observed object [7]. Rician-likelihood training for magnitude data is also established [8]. Neither self-supervision, learned Hankel processing nor the Rician loss is new on its own.

We recover a complex multiuser channel from noisy magnitudes and a known reference, rather than restoring measured complex samples. A small controller sets filtering strength instead of generating low-rank factors. Our companion manuscript studies classical Hankel estimation and rank selection [9]; here we learn filtering without channel labels.

Structured alternating least squares [10], data-driven Opt-Shrink [11], low-rank Hankel denoising (LRHD) [12] and robust Hankel completion [13] already provide other structural methods. Their original linear or direct matrix observations differ from our referenced magnitudes. Sparse amplitude-only beam training is also relevant [14], but uses adaptive sensing without our additive reference. We do not claim an optimal shrinker or superiority over these complete methods. The supplement distinguishes their assumptions and reports a limited post-processing comparison.

III. Receiver and Channel Model

An array with N elements receives P known training symbols, or pilots, from each of K users. The observations are

$$Z = |GS + B + W|, \qquad W_{np} \sim \mathcal{CN}(0, v). \tag{1}$$

Here $G \in \mathbb{C}^{N\times K}$ is the channel, $S \in \mathbb{C}^{K\times P}$ contains the pilots, and $B \in \mathbb{C}^{N\times P}$ is the known reference. Magnitudes act entrywise. Independent noise has total complex variance v, or $v/2$ in each real component. The reference makes magnitudes depend on relative signal phases, but does not ensure accurate recovery in every case. Reference calibration and noise variance are known, except in stated input-mismatch checks. Receiver conversion gain is absorbed into the scaling. This effective model does not simulate atomic dynamics or optical readout.

For the half-wavelength uniform linear array,

$$g_{nk} = \sum_{\ell=1}^{L_k} \alpha_{\ell k} e^{-j\pi n \sin\theta_{\ell k}}, \quad n = 0, \ldots, N-1. \tag{2}$$

Independent gains follow $\mathcal{CN}(0, 1/L_k)$, giving each user unit average channel power. Angles are uniform on $[-\pi/2, \pi/2]$; pilot entries follow $\mathcal{CN}(0, 1)$. The reference has a known spatial response and constant waveform. Define $\mathrm{SNR} =$

A. Haatim is with the Department of Electrical and Electronics Engineering, Birla Institute of Technology and Science, Pilani, India (e-mail: f20220519@pilani.bits-pilani.ac.in).

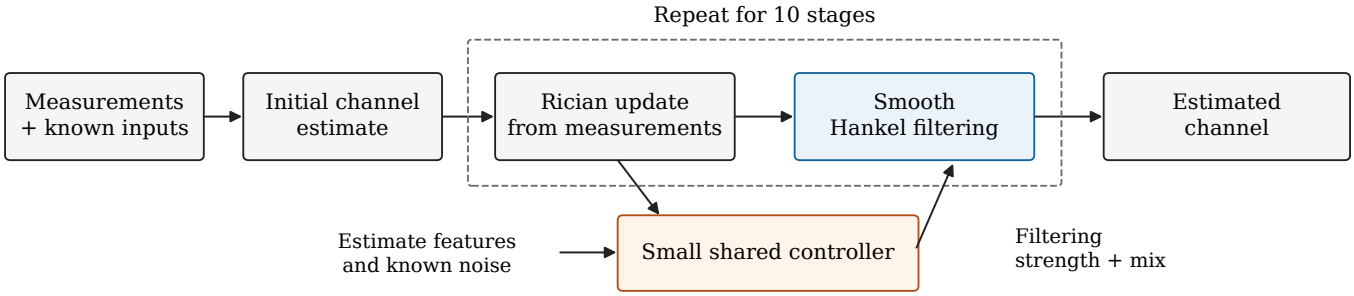


Fig. 1. Estimator overview. The controller chooses filtering strength and mixing. During training, nine withheld pilots score predictions but do not enter initialization or the ten stages.

$10\log_{10}(K/v)$ and $\mathrm{RSR} = 10\log_{10}|\alpha_b|^2$, where α_b is reference amplitude. RSR uses one user's power: for three users, RSR 12 dB is about 7.23 dB relative to their combined power.

A. What the Measurements Can Determine

For one array element, write $\mu_p = g^T s_p + b_p$ and $x = [\mathrm{Re}\, g^T, \mathrm{Im}\, g^T]^T$. If $\mu_p \neq 0$, the real Jacobian of $|\mu_p|$ has row

$$A_p = \left[\mathrm{Re}\, c_p^T, -\mathrm{Im}\, c_p^T\right], \qquad c_p = \mu_p^* s_p/|\mu_p|. \tag{3}$$

Full column rank of A ensures local recovery; a small minimum singular value means poor stability. Thus $P \geq 2K$ is necessary for this condition, not sufficient. Full row rank of S alone is insufficient. With $B = 0$, channel rotation leaves magnitudes unchanged. A known reference can break this ambiguity, but its strength alone does not ensure recovery. This unrestricted-row condition is local, not a global or sparse-channel guarantee [15]. Proof and information matrix are in the supplement.

IV. Proposed Estimator

The estimator starts with a rough channel guess. It repeats two steps ten times: one pulls the guess toward the measurements, and the other smooths it using the array structure. A small network decides how much smoothing to apply at each step (Fig. 1).

A. Measurement Update

The magnitude of a complex signal with Gaussian noise follows a Rician distribution. For noiseless magnitude $a = |\mu|$, its density is

$$p(z \mid a, v) = \frac{2z}{v} e^{-(z^2+a^2)/v} I_0(2za/v), \tag{4}$$

where I_0 is a modified Bessel function. Let $M = GS + B$ and $\kappa = 2Z|M|/v$. Entrywise, form

$$Q = Z\frac{I_1(\kappa)}{I_0(\kappa)}\frac{M}{|M|}, \tag{5}$$

using the continuous limit at $M = 0$. The expectation-maximization (EM) update and learned step are

$$G_{\mathrm{EM}} = (Q - B)S^H(SS^H)^{-1}, \tag{6}$$

$$G_{\mathrm{dc}} = G + \gamma_t(G_{\mathrm{EM}} - G), \quad \gamma_t = 2\,\mathrm{sigmoid}(a_t). \tag{7}$$

A linear solve evaluates the inverse. Input pilots must have full row rank. Ten stages are a fixed computation budget, not a proved optimum. Separately trained five- and twenty-stage models have not been compared.

B. Smooth Hankel Filtering

Each updated user channel forms a Hankel matrix $H = \mathcal{H}(g_{\mathrm{dc},k})$, with equal entries along each antidiagonal. At $N = 32$, it is 17×16. Singular values describe component strengths. Hard rank truncation deletes the smaller components. Our smooth rule is

$$\widehat{\sigma}_i = \frac{\sigma_i^3}{\sigma_i^2 + \tau^2}. \tag{8}$$

Small components shrink more without being deleted outright, but may contain real channel details.

Set $q = \lceil N/2 \rceil$, $s_H = \|H\|_F/\sqrt{q}$, and $\tau = s_H[0.001 + \mathrm{softplus}(u)]$. The equivalent operation is

$$\widehat{H} = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \tag{9}$$

$$g_k^{t+1} = g_{\mathrm{dc},k} + \eta_{t,k}(\mathcal{H}^\dagger \widehat{H}_k - g_{\mathrm{dc},k}), \tag{10}$$

where $\mathcal{H}^\dagger$ averages antidiagonals and $\eta_{t,k} = \mathrm{sigmoid}(w)$. The code normalizes by s_H before solving and guards zero scales. Gradients pass through the solve, without an identity-backward approximation. Neither control estimates path count. The filter does not increase the Hankel Frobenius norm, but this does not imply smaller channel error. The relaxed EM step decreases its quadratic surrogate for $0 \leq \gamma_t \leq 2$; the following filter need not decrease the likelihood. We make no convergence claim for the full estimator.

C. Controller Inputs

The network has eight inputs, 16 hidden units, two outputs and SiLU activation $x\,\mathrm{sigmoid}(x)$. Its 178 parameters and ten step sizes give 188. For $p_i = \sigma_i/\sum_j \sigma_j$, its first three inputs are

$$e^{-\sum_i p_i \log p_i}/q, \qquad p_1, \qquad \sum_{i=q-\lceil q/2 \rceil+1}^{q} p_i. \tag{11}$$

They tell the network how spread out the components are, how much the largest contributes, and how much the smaller components contribute. Two more inputs measure disagreement

TABLE I
MAIN SIMULATION AND TRAINING SETTINGS

Setting	Value
Array elements / users / pilots	32/3/30
Training / validation cases	16,384/2,048
SNR values (dB)	$-10, -5, 0, 5, 10, 15, 20$
Training RSR values (dB)	$5, 12, 20$
Training seeds	3, sharing training cases
Estimator stages / parameters	10/188
Training epochs / batch size	20/32
Optimizer / learning rate	Adam / 0.001
Input / withheld pilots	21/9
Training / evaluation precision	Single / double

with observations and expected noise relative to estimated channel power:

$$f_4 = \log\left(1 + \frac{\operatorname{mean}[(Z - |G_{\mathrm{dc}}S + B|)^2]}{v}\right), \tag{12}$$

$$f_{5,k} = \log\left(1 + \frac{v\,\operatorname{tr}[(SS^H)^{-1}]}{K\widehat{p}_k}\right), \quad \widehat{p}_k = \|g_{\mathrm{dc},k}\|^2/N. \tag{13}$$

Inputs $P/(2K)$, $\log q$ and $(t+1)/10$ give measurement count, size and stage ($t = 0, \ldots, 9$). Denominator guards avoid division by zero. True channels, path counts and family labels are not supplied. Weights stay fixed.

D. Training Without True Channels

Of 30 pilot columns, 21 enter the estimator and nine are withheld. Training scores predicted withheld magnitudes using the Rician loss

$$\ell(z, \mu, v) = \frac{(z - |\mu|)^2}{v} - \log I_{0e}(2z|\mu|/v). \tag{14}$$

Here I_{0e} is the scaled Bessel function. We omit an observation-only term and use small-field series at continuous limits. The loss fits Rician amplitude, not its mean. Splits change each epoch; separate validation cases with a fixed split select the checkpoint. Deployment uses all pilots, without extra measurements or weight training. This avoids test-driven selection; it does not make withheld likelihood equivalent to channel NMSE or remove dependence on the validation mixture.

V. SIMULATION STUDY

A. Setup and Comparators

Table I gives the settings. Half the training cases have three to seven paths per user; the rest use fixed counts of 2, 4, 8, 12 and 16. Validation weights 126 SNR, RSR and path-setting combinations equally. These condition labels set validation weights only; model selection uses withheld likelihood, not channel error.

The main study trains 18 models: two updates, three filtering choices and three seeds. Seeds change weights, batch order and pilot splits, but share training cases. Training took 3.67 hours with overlapping workers, excluding data preparation and preflight. Tests cover 32 conditions with 400 paired cases each. Two noise-input tests reuse cases, giving 12,000 distinct cases. All methods share channels, pilots and noise per case.

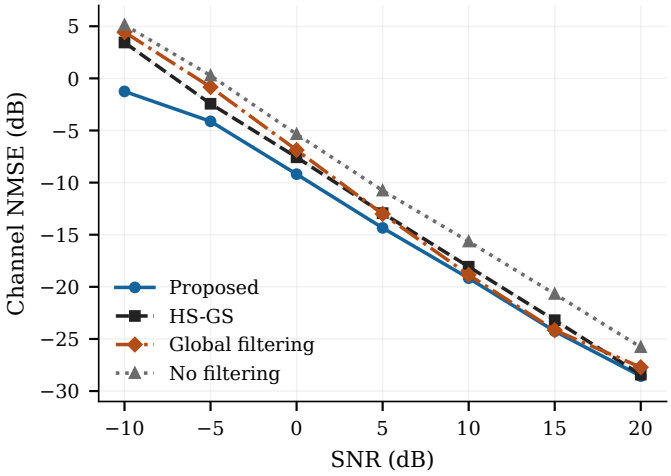


Fig. 2. Main SNR test: $N = 32$, $K = 3$, $P = 30$, RSR 12 dB, and three-to-seven-path channels. Lines average per-seed scores; bands are conditional paired-case 95% intervals.

HS-GS, the companion study's classical Hankel estimator [9], fits ranks $1, \ldots, q$ on 70% of pilots, using 25 EM updates and four rank/Hankel projection sweeps per update. Squared magnitude residuals on the remaining pilots select a rank. It then refits all pilots for 100 updates. Rank search uses no extra observations and counts toward time. Its rank-selection loss differs from our Rician training loss. Other controls use EM, Rician or amplitude-gradient updates. Global filtering has 30 parameters and scales its threshold with s_H; no filtering has ten. Hard-Hankel Rician uses development-selected rank ten and one projection sweep per update. Constant-output controls replace adaptive outputs with validation averages without retraining.

Our earlier URformer-style model has 1,586,900 parameters and used 80,000 labelled examples, fixed pilots, $P = 20$ and RSR 10 dB. It has no matched evaluation under this paper's protocol; its earlier scores are not included in Table II.

Channel error is normalized mean squared error (NMSE):

$$\mathrm{NMSE}_{\mathrm{dB}} = 10 \log_{10} \frac{\sum_i \|\widehat{G}_i - G_i\|_F^2}{\sum_i \|G_i\|_F^2}. \tag{15}$$

Lower is better. Per-seed dB scores and the seven main SNR settings are averaged equally. Positive gain means lower error. A severe error has more energy than the true channel (per-case linear NMSE above one). Pointwise 95% intervals use 4,000 paired whole-case resamples, conditional on saved models, excluding training-dataset uncertainty. Classical stopping methods have a 1,000-update cap; reaching it need not mean convergence.

B. Accuracy and Processing Time

HS-GS is the strongest original classical comparator on the declared average (Table II). The adaptive gradient-descent (GD) core has lower test NMSE than the primary EM core. We retain EM because validation likelihood selected it before test generation.

TABLE II
ALL MAIN-STUDY COMPARATORS

Method	NMSE (dB)	Time (ms)	Severe (%)
EM adaptive (primary)	-14.42	8.24	0.45
EM learned steps only	-10.41	2.98	22.11
EM global filtering	-12.43	5.61	19.19
EM constant-output control	-9.19	5.61	11.42
GD adaptive (secondary)	-14.59	8.00	0.85
GD learned steps only	-10.44	2.73	21.15
GD global filtering	-12.57	5.38	18.37
GD constant-output control	-9.95	5.36	12.36
HS-GS, rank search	-12.74	525.62	14.25
EM, 10 updates	-10.23	4.10	20.64
EM, 100 updates	-10.28	17.01	23.57
Rician GD, 10 updates	-9.31	4.17	17.86
Rician GD, 100 updates	-10.29	17.40	23.50
Rician GD, stopping rule	-10.27	45.37	23.75
Amplitude GD, 10 updates	-8.80	3.29	20.21
Amplitude GD, 100 updates	-9.65	8.87	26.96
Amplitude GD, stopping rule	-9.61	20.93	27.11
Rician GD, hard Hankel	-11.06	47.39	20.93
Linearized least squares	-6.92	1.06	19.86

NMSE and severe-error rates average seven primary SNR conditions equally. Time is the median at SNR 5 dB, RSR 12 dB. Learned scores average three seeds; classical cases are not counted three times.

The EM model's average gain over HS-GS is 1.675 dB, with interval $[1.618, 1.730]$ dB and seed-specific gains of 1.533–1.787 dB (Fig. 2). At 20 dB, the gain is 0.139 dB with interval $[-0.017, 0.294]$ dB, which does not establish a clear improvement.

Severe errors fall from 14.25% for HS-GS to 0.45% for the proposed method. In the tested SNR mixture, that is roughly one badly wrong estimate in seven cases versus one in 220.

Median time is 8.24 ms versus 525.62 ms for HS-GS, about 64 times faster (Table II). Timing includes initialization and rank search, using one Apple M2 CPU thread, double precision, batch size one, 100 cases and three repeats at SNR 5 dB and RSR 12 dB. These are implementation-specific measurements: learned methods use PyTorch and classical methods use NumPy/SciPy. Offline training is separate. No filtering is faster but less accurate. With $m = N - q + 1$, one adaptive stage costs $O(NKP + NK^2 + K^3 + Kmq^2 + Kq^3)$: the spectral features still require an SVD. HS-GS repeats comparable projections across candidate ranks and updates. The supplement separates initialization, update and projection costs; 188 weights do not imply negligible computation.

C. Operating Limits

The proposed method loses 1.617 dB at RSR 0 dB, 1.104 dB with two paths, and 0.203 dB with three paths (Fig. 3). Good average performance does not ensure good performance with a weak reference or a simple channel.

Transfer tests give gains for noise inputs wrong by ± 3 dB and three channel alternatives: four clusters of ten rays within $\pm 5^\circ$, ten equal-power paths, or ten paths with geometrically decreasing powers. Alternatives match base-channel energy and are tested only at SNR 5 dB, RSR 12 dB. Dimension tests use $N \in \{8, 16, 32, 64\}$, $K \in \{2, 3, 4\}$ and $P \in \{10, 20, 30\}$, changing one at a time. At $N = 16$ and 64, gains are 1.223 and 0.165 dB; at $N = 8$ the interval includes zero. No tested pilot budget approaches $2K$ closely. The supplement reports all dimension changes. Path counts overlap training; these tests do not resolve failures under earlier protocols.

D. Further Comparisons

A follow-up trains 15 models and uses two fresh test sets, each with seven SNR settings and two difficult settings of 200 cases. Six primary comparisons are adjusted for multiple testing per set. Independent training-data runs gain 1.939 and 1.837 dB over HS-GS; each dataset uses one seed, limiting replication. These are separate from the main 1.675 dB result. The supplement retains all six comparisons.

A matched-architecture supervised recipe uses mean linear NMSE and the same input pilots and training budget. Its approximately 1 dB gap depends on checkpoint and averaging choices: it reduces to about 0.73 or 0.36 dB under alternative rules. Supervision wins in some settings, including two paths. This is not a general superiority claim for self-supervision.

Repeated filtering gains about 0.70 dB over separately trained final-only filtering. A separate timing sample gives 4.65 versus 2.05 ms. Mixing is usually near one; fixing it at one gives a small average gain but worsens two of three seeds on two-path cases. The controller adjusts shrinkage strength, not a reliable on/off decision.

E. Which Inputs Help?

We compare full, spectrum-only, noise/residual-only and size/stage-only controllers. Size and stage remain in every version. All use the same update and filter, trained from scratch on 4,096 shared cases, with three seeds and 12 epochs. Two fresh sets contain 2,304 cases: seven SNR settings and two difficult settings, 128 cases per condition per set.

Full inputs gain 1.22–1.25 dB over spectrum-only, 0.57–0.58 dB over noise/residual-only, and 2.21–2.23 dB over size/stage-only inputs; adjusted intervals remain positive. Yet they lose 0.39–0.40 dB against noise/residual-only at SNR 5 dB, failing the declared 0.25 dB protection requirement, and 0.82–0.93 dB on two-path cases. The supplement retains the figure. Both groups help the average, not every channel. All variants store 188 parameters with masked weights inactive; noise variance still enters the measurement update and channel information still enters the filter.

F. Small Reviewer Diagnostics

Fresh checks use 32 cases per cell and frozen weights, not a new confirmation study. Rician rank scoring improves HS-GS by 0.182 dB; the learned model retains a 1.282 dB average gain. A separately development-tuned EM100-plus-OptShrink control narrows the learned gain to 0.552 dB and wins at SNR 20 dB and RSR 0 dB. In matched one-pass post-processing, OptShrink beats cubic filtering by 2.042 dB. Thus the cubic rule itself is not superior. Supplying noise variance 6 dB too high also makes the learned model 1.409 dB worse than Rician-ranked HS-GS. The supplement reports paired intervals, reference-input errors and controller trajectories. These results strengthen the comparisons but narrow the useful operating range.

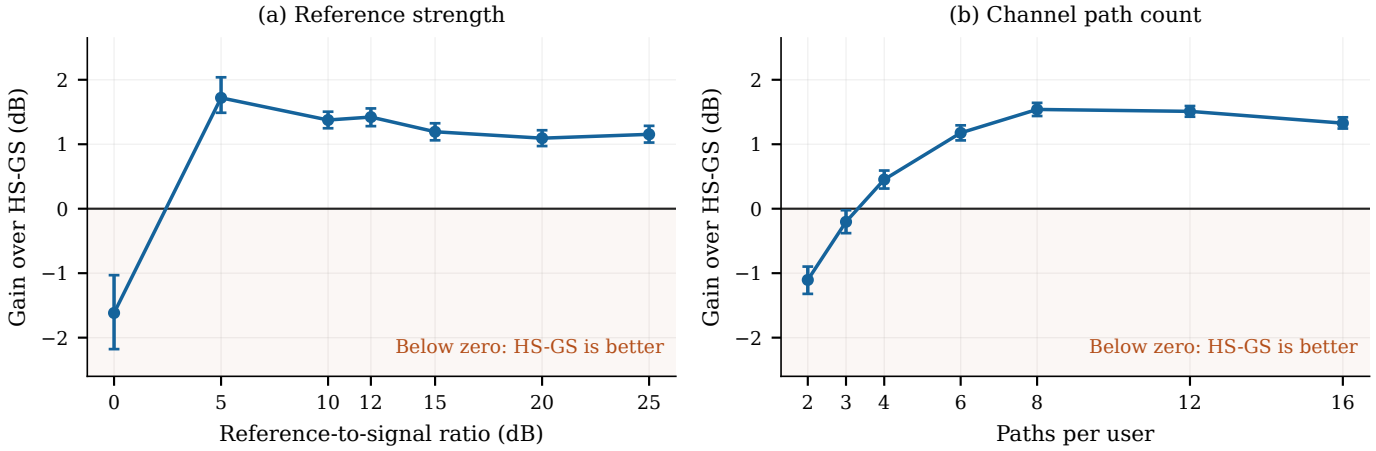


Fig. 3. Limits at SNR 5 dB: (a) reference strength with three-to-seven-path channels; (b) fixed path counts at RSR 12 dB. Positive gain favors the proposed estimator. Bars are conditional pointwise paired-case 95% intervals. Path counts overlap training.

VI. Discussion and Conclusion

A small model can learn useful Hankel filtering without true-channel targets. Its practical setting is repeated estimation near the simulated training population, where offline training is reused. The average accuracy and time gains come with weak-reference and few-path losses.

All results are simulations. Reference calibration is assumed known, and no physical receiver measurements are used. Three seeds sharing training cases are not three independent datasets. We have not proved unique channel recovery in every setting or convergence of the full learned estimator. The comparisons also do not establish superiority over published URformer performance. Noise-dependent regularized classical controls and broader independent training/pilot replications would be needed for a stronger robustness claim. We do not infer the cause of a loss solely from its association with SNR, path count or reference strength. Two-path and RSR 5 dB examples already occur in training, but RSR 0 dB does not. Changing their weights or adding a curriculum has not been tested. No robustness to physical shot statistics, unknown reference drift or two-dimensional arrays is established. A block-Hankel extension would require a separate geometry model and validation, not just reshaping the input.

The supplement records initialization, numerical safeguards, controller trajectories and bounded reviewer diagnostics. These exploratory checks do not replace the original frozen confirmation results. The original reproducibility package contains compact outputs and a replay subset, not every observation/prediction array.

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